#ManyDaughters Pre-Analysis Plan

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1 Introduction

This pre-analysis plan outlines our contribution to the #ManyDaughters project, which examines how having daughters influences attitudes, preferences, and behaviors related to gender equality. Using data from the German Socio-Economic Panel (SOEP) from 2000 to 2020, we analyze four hypotheses:

1. **H1:** Having daughters leads to stronger support for gender equality (as measured by attitudes).
2. **H2:** Having daughters leads to more left-leaning political preferences.
3. **H3:** Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
4. **H4:** Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

We exploit the random assignment of children's gender for causal identification and employ panel regressions with respondent and time-fixed effects, using three key outcome variables:

- a gender equality attitude index (2016–2018, H1 and H4),

- left–right political scores (based on party support, H2),
- household labor equity (time-use data, H3).

The sample includes respondents with clearly identified genders and children. Statistical inference uses one-sided t-tests ($p < 0.05$) with household-level clustering.

2 Sample and inclusion criteria

We test the hypotheses defined by the project using SOEP data from the years 2000 to 2020 in SOEP-Core (in particular, the datasets "pl", "hl", "biol", "biobirth", "bioparen", "pgen", "hgen", and "ppathl").

We include observations from all respondents in the SOEP-Core dataset between 2000 and 2020 if:

1. the respondent has, in at least one year of the sample period, at least one biological child (`biobirth/sumkids` > 0 in `bioyear` $\in [2000, 2020]$),
2. the respondent's gender is clearly identified as either male or female, and the same during the entire sample period (to facilitate the test of gender-specific hypotheses, using `biobirth/sex`), and
3. the gender of all the respondents' biological children is identified as either male or female (to clearly differentiate the effect of having a daughter compared to having a son, using `biobirth/kidsex01`–`biobirth/kidsex19`).

3 Survey questions

3.1 Survey questions on the timing of the birth and the gender of the children

The SOEP contains up to 19 children for every respondent. For all respondents in our sample, we construct two variables for each year: `has_daughter` and `has_child`. `has_child` is set to 1 starting in the year their first child was born (`biobirth/kidgeb01`) and 0 otherwise. `has_daughter` is set to 1 starting in the year their first daughter was born (based on `biobirth/kidgeb[year]` for the first [year] where `biobirth/kidsex[year]` indicates female).

As we use the random assignment of children's gender for causal identification, we only include biological children in the two variables above. We do not exclude in vitro fertilization, as in Germany, according to the Embryo Protection Act (Embryonenschutzgesetz; ESchG), in vitro fertilization embryos cannot be selected by gender.

3.2 Survey questions on outcome variables

3.2.1 Support for gender equality

The outcome variable measuring support for gender equality is an aggregated index constructed from three survey items. The index includes responses regarding decision-making authority within the household ("At home, the husband should have the final say," `plj0625`), educational priorities between sons and daughters ("For parents, vocational training or higher education for their sons should be more important," `plj0624`), and attitudes toward married women's financial independence ("Even a married woman should have a paid job," `plj0622`).[1] Each item contributes equally to the index, which is then normalized to values between 0 and 1, with higher values indicating greater support for gender equality. Unfortunately, items asking for attitudes

[1] We do not include the following survey items to the index: "Women have the same rights as men" (`plj0620`); "Having a job is the best way for a woman to be independent" (`plj0621`); and "If a woman earns more money than her partner, this inevitably leads to problems" (`plj0623`), as responses may reflect perceptions of societal problems or attitudes rather than personal support for gender equality.

about gender equality are only present between 2016 and 2018. Therefore, Hypotheses 1 and 4 are analyzed using data from 2016–2018 only.

3.2.2 Political preferences

Respondents are asked whether they support a political party in the SOEP variable `plh0011_h`. If they answer "yes," they are then asked which party they support (`plh0012_h`).

We use these two variables to construct a measure of political preferences along the left-right spectrum as follows:

1. For respondents who support a specific party, we assign the Left–Right Ideology average score from the Chapel Hill Expert Survey (CHES) for the corresponding party and period (Jolly et al, 2022). The CHES scores range from 0 (left) to 10 (right), vary over time, and are available from 1999 to 2019. As the CHES is not updated annually, we use the most recent available wave, accounting for the survey's publication lag. To align with the hypothesis, we invert the variable so that higher values indicate support for more left-leaning parties.

2. Respondents who do not indicate support for a specific party are assigned a value of 5. This value lies between the average CHES scores of the Social Democratic Party (SPD; center-left) and the Christian Democratic Union (CDU; center-right) in all survey waves, and is labeled as "center" in the CHES survey.

3.2.3 Gender equality in the household

For this hypothesis, we restrict the sample to respondents whose partner lives in the same household (`pld0133`) and has a different gender (`[individual]gen/partnr[year]`).

The SOEP includes questions on respondents' time use for household-related activities. The

following variables capture time spent on specific tasks:

- Hours spent running errands: `pli0040` (weekdays), `pli0054` (Saturdays), and `pli0011` (Sundays);
- Hours spent on housework: `pli0043_h` (weekdays), `pli0012_h` (Saturdays), `pli0016_h` (Sundays);
- Hours spent on childcare: `pli0044_h` (weekdays), `pli0019_h` (Saturdays), `pli0022_h` (Sundays);
- Hours spent supporting persons in need of care: `pli0046` (weekdays), `pli0055` (Saturdays), `pli0057` (Sundays);
- Hours spent on repairs: `pli0049_h` (weekdays), `pli0031_h` (Saturdays), `pli0034_v4` (Sundays).

For each respondent, we compute the total weekly hours spent on these activities, weighting weekdays by 5 and Saturdays and Sundays by 1. As household-related activities on Saturdays and Sundays are not collected yearly, we do not calculate total weekly hours unless all relevant variables are available.

To construct a measure of gender equality in household labor for each household and year, we calculate:

$$\text{Equity} = 1 - \frac{|h_f - h_m|}{h_f + h_m} \tag{1}$$

In Eq. (1), h_f (h_m) refers to the total weekly hours spent on these activities in a given year by the female (male) partner. The measure ranges from 0 (complete inequality) to 1 (perfect equality).

3.3 Control variables

We use the following control variables in our analysis:

- respondent's age (`biobirth/bioyear` – `biobirth/gebjahr`),
- respondent's age squared,
- respondent's number of sons (`biobirth/kidsex01` - `kidsex19`), and
- respondent's number of daughters (`biobirth/kidsex01` - `kidsex19`).

4 Hypotheses and tests

We use our analysis sample to examine the first three hypotheses using a standard fixed-effect panel regression with two dummy variables:

$$Y_{i,t} = \alpha_i + \gamma_t + \beta_1 \, B_{i,t} + \beta_2 \, D_{i,t} + X'_{i,t} \, \delta + \varepsilon_{i,t} \tag{2}$$

In Eq. (2), $Y_{i,t}$ indicates the outcome variable (specified below), with i designating the respondent and t the time unit (the year of the survey). α_i is the individual-fixed effect, controlling for all unobserved, time-invariant characteristics. γ_t captures time-fixed effects. $B_{i,t}$ is the dummy variable that turns from 0 to 1 with the first birth (`has_child`, as defined in Section 3.1). $D_{i,t}$ is a dummy variable that turns from 0 to 1 with the birth of the first daughter (`has_daughter`, as defined in Section 3.1). Thus, β_2 is the coefficient we are interested in for Hypotheses 1–3). $X_{i,t}$ is a vector of control variables, as specified in Section 3.3. Finally, $\varepsilon_{i,t}$ is the error term.

By employing this fixed-effects regression approach, we exploit the within-respondent variation to identify the effect of having a daughter (relative to having only sons). Specifically, identification relies on households experiencing temporal changes in their child composition.

While the respondents can choose when to have children, the children's gender is randomly determined. We distinguish the following types of respondents in our panel dataset:

- Respondents without variation in either dummy: for them, the birth and daughter dummies remain constant. They serve as baseline units but do not directly contribute to identifying the effects of interest.

- Respondents initially (upon entering the survey or the beginning of the sample period) having children (birth dummy = 1) but no daughters (daughter dummy = 0), later having a daughter: They provide clear within-respondent variation in the daughter dummy, identifying the incremental effect of having a daughter.

- Respondents simultaneously experiencing first childbirth and the birth of a daughter: For them, the birth and daughter effects occur concurrently; the two effects happen at the same time.

- Respondents experiencing first childbirth with only sons (birth dummy turns from 0 to 1, daughter dummy remains 0): They help directly identify the general effect of having children (without the daughter-specific effect).

- Respondents initially experiencing first childbirth with sons, later followed by a daughter: They offer the most robust within-respondent variation, separating the childbirth effect from the incremental effect of subsequently having a daughter.

We cluster at the household level and include a set of control variables (as specified in Section 3.3): the respondent's gender (male or female), their age and age squared, and the respondent's number of sons and number of daughters.

H1: Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality, (X for

respondents with daughters) > (X for respondents without daughters).

We test this hypothesis by using the outcome variable defined in Section 3.2.1 as the measure of respondents' support for gender equality in Eq. (2). We then assess whether the coefficient β_2 is significantly greater than zero using a one-sided t-test, interpreting a p-value below 0.05 as statistically significant evidence.

H2: Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).

We test this hypothesis by using the outcome variable defined in Section 3.2.2 as the measure of respondents' left-leaning political preferences in Eq. (2). We then assess whether the coefficient β_2 is significantly greater than zero using a one-sided t-test, interpreting a p-value below 0.05 as statistically significant evidence.

H3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters) > (Z for respondents without daughters).

We test this hypothesis by using the outcome variable defined in Section 3.2.3 as the measure of respondents' gender equality in the household in Eq. (2). We then assess whether the coefficient β_2 is significantly greater than zero using a one-sided t-test, interpreting a p-value below 0.05 as statistically significant evidence.

H4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality, (X for male respondents with daughters - X for male respondents without daughters)>(X for female respondents with daughters - X for female

respondents without daughters).

As Hypothesis 1, we test this hypothesis by using the variable defined in Section 3.2.1 as the measure of respondents' support for gender equality.

Due to the question of how the respondents' gender affects the impact of having a daughter on the support for gender equality, we have to extend the model in Eq. (2) by interaction effects for the respondents' gender:

$$Y_{i,t} = \alpha_i + \gamma_t + \beta_1 \text{ Male}_i + \beta_2 \, B_{i,t} + \beta_3 \, (\text{Male} \times B)_{i,t} + \beta_4 \, D_{i,t} + \beta_5 \, (\text{Male} \times D)_{i,t} + X'_{i,t} \, \delta + \varepsilon_{i,t} \quad (3)$$

In Eq. (3), in addition to Eq. (2), both the first-birth dummy variable (`has_child`) and the daughter dummy variable (`has_daughter`) are extended by an interaction term with the (male) gender of the respondent. We also include the male dummy variable to avoid that it is absorbed by the respondent-fixed effect α_i. β_1 measures men's support for gender equality (in contrast to women's). β_2 captures the effect of having children on support for gender equality. β_3 captures the effect of having children on support for gender equality for men. β_4 captures the effect of having a daughter on support for gender equality. β_5 captures the effect of having a daughter on support for gender equality for men.

Again, as in the tests of Hypotheses 1–3, we cluster at the household level and include control variables as outlined in Section 3.3.

We test Hypothesis 4 by assessing whether the coefficient β_5 is significantly greater than zero using a one-sided t-test, interpreting a p-value below 0.05 as statistically significant evidence.

References

Jolly S, Bakker R, Hooghe L, Marks G, Polk J, Rovny J, Steenbergen M, Vachudova MA (2022) Chapel hill expert survey trend file, 1999–2019. Electoral Studies 75:102,420